

# TrES-2b

R-1629

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_TRES-2\_150603 · created 2026-08-02T05:29:37+00:00

BENCHMARK: ACCURATE — fit consistent with published values

## Results

|                                                   |                                   |
|---------------------------------------------------|-----------------------------------|
| Mid-Transit Time (Tmid)                           | 2457176.84537 +/- 0.00036 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.141 +/- 0.014                   |
| Transit depth (Rp/Rs)^2                           | 2.0 +/- 0.4 [%]                   |
| Orbital Inclination (inc)                         | 83.35 +/- 0.27                    |
| Airmass coefficient 1 (a1)                        | 1.1774 +/- 0.0079                 |
| Airmass coefficient 2 (a2)                        | -0.15 +/- 0.012                   |
| Scatter in the residuals of the lightcurve fit is | 0.67 %                            |
| Best Comparison Star                              | None                              |
| Optimal Aperture                                  | 3.95                              |
| Optimal Annulus                                   | 13.94                             |
| Transit Duration (day)                            | 0.0681 +/- 0.0054                 |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.141 ± 0.014 vs 0.1254 (deviation 0.82σ) |
| Duration fit vs published | 0.0681 d vs 0.0746 d (deviation 0.88σ)    |
| Mid-time O–C              | 0.9 min                                   |
| Depth SNR                 | 7.4                                       |
| Residual scatter          | 0.67 %                                    |

## Light curve

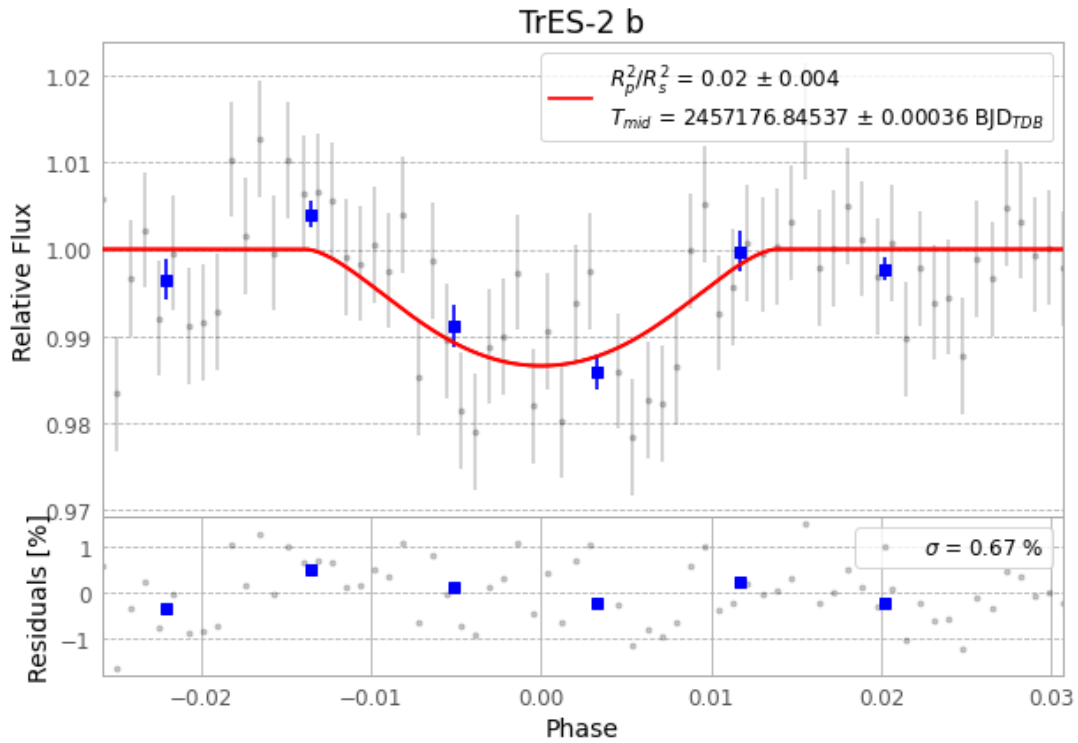


Figure 1 — FinalLightCurve\_TrES-2 b\_2015-06-02.png (SHA-256 cbb9d9736207a50a...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [190, 234], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 69d89c2add8e0b3b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

68 FITS frames (45 MB), TRES-2150603064212.FITS → TRES-2150603100312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-150603.log (e6d3a2b1de8d...), FinalLightCurve\_TrES-2 b\_2015-06-02.png (cbb9d9736207...), FinalLightCurve\_TrES-2 b\_2015-06-02.csv (dc08f77981be...)

## Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 05:29Z.